# Constructions With Relations

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This chapter contains some material about constructions with relations. Notably, we discuss and explore:

1. The existence or non-existence of Kan extensions and Kan lifts in the 2-category **Rel** (Section 9.2).
2. The various kinds of constructions involving relations, such as graphs, domains, ranges, unions, intersections, products, converse relations, composition of relations, and collages (Section 9.3).

## Contents

## 9.1 Co/Limits in the Category of Relations

This section is currently just a stub, and will be properly developed later on.

## 9.2 Kan Extensions and Kan Lifts in the 2-Category of Relations

### 9.2.1 Left Kan Extensions in **Rel**

**Proposition 9.2.1.1.1 ▶ Left Kan Extensions in Rel**

Let $R\colon A \nrightarrow B$ be a relation.

1. *Non-Existence of All Left Kan Extensions in* **Rel**. Not all relations in **Rel** admit left Kan extensions.

2. *Characterisation of Relations Admitting Left Kan Extensions Along Them.* The following conditions are equivalent:

   (a) The left Kan extension

   $$\mathrm{Lan}_R\colon \mathbf{Rel}(A, X) \to \mathbf{Rel}(B, X)$$

   along $R$ exists.

   (b) The relation $R$ admits a left adjoint in **Rel**.

   (c) The relation $R$ is of the form $f^{-1}$ (as in **??**) for some function $f$.

**PROOF 9.2.1.1.2 ▶ PROOF OF PROPOSITION 9.2.1.1.1**

Item 1: Non-Existence of All Left Kan Extensions in **Rel**

Omitted, but will eventually follow Fosco Loregian's comment on [**MO460656**].

Item 2: Characterisation of Relations Admitting Left Kan Extensions

Omitted, but will eventually follow Tim Campion's answer to to [**MO460656**].

**QUESTION 9.2.1.1.3 ▶ EXISTENCE OF SPECIFIC LEFT KAN EXTENSIONS OF RELATIONS**

Given relations $S\colon A \nrightarrow X$ and $R\colon A \nrightarrow B$, is there a characterisation of when the left Kan extension

$$\mathrm{Lan}_S(R)\colon B \nrightarrow X$$

exists in terms of properties of $R$ and $S$?
This question also appears as [**MO461592**].

**QUESTION 9.2.1.1.4 ▶ EXPLICIT DESCRIPTION OF LEFT KAN EXTENSIONS ALONG FUNCTIONS**

As shown in Item 2 of Proposition 9.2.1.1.1, the left Kan extension

$$\mathrm{Lan}_R\colon \mathbf{Rel}(A, X) \to \mathbf{Rel}(B, X)$$

along a relation of the form $R = f^{-1}$ exists. Is there a explicit description of it, similarly to the explicit description of right Kan extensions given

in Proposition 9.2.3.1.1?
This question also appears as [**MO461592**].

### 9.2.2 Left Kan Lifts in Rel

**Proposition 9.2.2.1.1 ▶ Left Kan Lifts in Rel**

Let $R\colon A \nrightarrow B$ be a relation.

1. *Non-Existence of All Left Kan Lifts in* **Rel**. Not all relations in **Rel** admit left Kan lifts.

2. *Characterisation of Relations Admitting Left Kan Lifts Along Them.* The following conditions are equivalent:

   (a) The left Kan lift

   $$\mathrm{Lift}_R\colon \mathbf{Rel}(X, B) \to \mathbf{Rel}(X, A)$$

   along $R$ exists.

   (b) The relation $R$ admits a right adjoint in **Rel**.

   (c) The relation $R$ is of the form $\mathrm{Gr}(f)$ (as in **??**) for some function $f$.

**Proof 9.2.2.1.2 ▶ Proof of Proposition 9.2.2.1.1**

Item 1: Non-Existence of All Left Kan Lifts in **Rel**

Omitted, but will eventually follow (the dual of) Fosco Loregian's comment on [**MO460656**].

Item 2: Characterisation of Relations Admitting Left Kan Lifts Along

Omitted, but will eventually follow Tim Campion's answer to to [**MO460656**].

**Question 9.2.2.1.3 ▶ Existence of Specific Left Kan Lifts of Relations**

Given relations $S\colon A \nrightarrow X$ and $R\colon A \nrightarrow B$, is there a characterisation of when the left Kan lift

$$\mathrm{Lift}_S(R)\colon X \nrightarrow A$$

exists in terms of properties of $R$ and $S$?
This question also appears as [**MO461592**].

**Question 9.2.2.1.4 ▶ Explicit Description of Left Kan Lifts Along Functions**

As shown in Item 2 of Proposition 9.2.2.1.1, the left Kan lift

$$\mathrm{Lift}_R\colon \mathbf{Rel}(X, B) \to \mathbf{Rel}(X, A)$$

along a relation of the form $R = \mathrm{Gr}(f)$ exists. Is there a explicit description of it, similarly to the explicit description of right Kan lifts given in Proposition 9.2.4.1.1?
This question also appears as [**MO461592**].

## 9.2.3 Right Kan Extensions in Rel

Let $R\colon A \nrightarrow B$ be a relation.

**Proposition 9.2.3.1.1 ▶ Existence of Right Kan Extensions in Rel**

The right Kan extension

$$\mathrm{Ran}_R\colon \mathrm{Rel}(A, X) \to \mathrm{Rel}(B, X)$$

along $R$ in **Rel** exists and is given by

$$\mathrm{Ran}_R(S) \stackrel{\text{def}}{=} \int_{a \in A} \mathbf{Hom}_{\{\mathrm{t},\mathrm{f}\}}\left(R_a^{-2}, S_a^{-1}\right)$$

for each $S \in \mathrm{Rel}(A, X)$, so that the following conditions are equivalent:

1. We have $b \sim_{\mathrm{Ran}_R(S)} x$.
2. For each $a \in A$, if $a \sim_R b$, then $a \sim_S x$.

**Proof 9.2.3.1.2 ▶ Proof of Proposition 9.2.3.1.1**

We have

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{Rel}(A,X)}(S \diamond R, T) &\cong \int_{a\in A}\int_{x\in X} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}((S\diamond R)^x_a, T^x_a) \\
&\cong \int_{a\in A}\int_{x\in X} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(\left(\int^{b\in B} S^x_b \times R^b_a\right), T^x_a\right) \\
&\cong \int_{a\in A}\int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(S^x_b \times R^b_a, T^x_a\right) \\
&\cong \int_{a\in A}\int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(S^x_b, \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_a, T^x_a\right)\right) \\
&\cong \int_{b\in B}\int_{x\in X}\int_{a\in A} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(S^x_b, \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_a, T^x_a\right)\right) \\
&\cong \int_{b\in B}\int_{x\in X} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(S^x_b, \int_{a\in A} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_a, T^x_a\right)\right) \\
&\cong \mathrm{Hom}_{\mathbf{Rel}(B,X)}\left(S, \int_{a\in A} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}(R^{-2}_a, T^{-1}_a)\right)
\end{aligned}$$

naturally in each $S \in \mathbf{Rel}(B,X)$ and each $T \in \mathbf{Rel}(A,X)$, showing that

$$\int_{a\in A} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^{-2}_a, T^{-1}_a\right)$$

is right adjoint to the precomposition functor $-\diamond R$, being thus the right Kan extension along $R$. Here we have used the following results, respectively (i.e. for each $\cong$ sign):

1. Relations, Item 1 of Proposition 8.1.1.1.10.
2. **??**.
3. Ends and Coends, **??** of **??**.
4. Sets, Proposition 3.2.2.1.5.
5. Ends and Coends, **??** of **??**.
6. Ends and Coends, **??** of **??**.
7. Relations, Item 1 of Proposition 8.1.1.1.10.

This finishes the proof. ∎

### 9.2.4 Right Kan Lifts in Rel

Let $R\colon A \nrightarrow B$ be a relation.

**Proposition 9.2.4.1.1 ▶ Existence of Right Kan Lifts in Rel**

The right Kan lift

$$\mathrm{Rift}_R\colon \mathrm{Rel}(X,B) \to \mathrm{Rel}(X,A)$$

along $R$ in **Rel** exists and is given by

$$\mathrm{Rift}_R(S) \overset{\text{def}}{=} \int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_{-_1}, S^b_{-_2}\right)$$

for each $S \in \mathrm{Rel}(X,B)$, so that the following conditions are equivalent:

1. We have $x \sim_{\mathrm{Rift}_R(S)} a$.
2. For each $b \in B$, if $a \sim_R b$, then $x \sim_S b$.

**Proof 9.2.4.1.2 ▶ Proof of Proposition 9.2.4.1.1**

We have

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{Rel}(X,B)}(R \diamond S, T) &\cong \int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left((R\diamond S)^b_x, T^b_x\right)\\
&\cong \int_{x\in X}\int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(\left(\int^{a\in A} R^b_a \times S^a_x\right), T^b_x\right)\\
&\cong \int_{x\in X}\int_{b\in B}\int_{a\in A} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_a \times S^a_x, T^b_x\right)\\
&\cong \int_{x\in X}\int_{b\in B}\int_{a\in A} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(S^a_x, \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_a, T^b_x\right)\right)\\
&\cong \int_{x\in X}\int_{a\in A}\int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(S^a_x, \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_a, T^b_x\right)\right)\\
&\cong \int_{x\in X}\int_{a\in A} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(S^a_x, \int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_a, T^b_x\right)\right)\\
&\cong \mathrm{Hom}_{\mathbf{Rel}(X,A)}\left(S, \int_{b\in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_{-_1}, T^b_{-_2}\right)\right)
\end{aligned}$$

naturally in each $S \in \mathbf{Rel}(X, A)$ and each $T \in \mathbf{Rel}(X, B)$, showing that

$$\int_{b \in B} \mathbf{Hom}_{\{\mathsf{t},\mathsf{f}\}}\left(R^b_{-_1}, S^b_{-_2}\right)$$

is right adjoint to the postcomposition functor $R \diamond -$, being thus the right Kan lift along $R$. Here we have used the following results, respectively (i.e. for each $\cong$ sign):

1. Relations, Item 1 of Proposition 8.1.1.1.10.
2. **??**.
3. Ends and Coends, **??** of **??**.
4. Sets, Proposition 3.2.2.1.5.
5. Ends and Coends, **??** of **??**.
6. Ends and Coends, **??** of **??**.
7. Relations, Item 1 of Proposition 8.1.1.1.10.

This finishes the proof. ∎

## 9.3 More Constructions With Relations

### 9.3.1 The Domain and Range of a Relation

Let $A$ and $B$ be sets.

**Definition 9.3.1.1.1 ▶ The Domain and Range of a Relation**

Let $R \subset A \times B$ be a relation.[1,2]

1. The **domain of** $R$ is the subset $\mathrm{dom}(R)$ of $A$ defined by

$$\mathrm{dom}(R) \stackrel{\text{def}}{=} \left\{ a \in A \;\middle|\; \begin{array}{l} \text{there exists some } b \in B \\ \text{such that } a \sim_R b \end{array} \right\}.$$

2. The **range of** $R$ is the subset $\mathrm{range}(R)$ of $B$ defined by

$$\mathrm{range}(R) \stackrel{\text{def}}{=} \left\{ b \in B \;\middle|\; \begin{array}{l} \text{there exists some } a \in A \\ \text{such that } a \sim_R b \end{array} \right\}.$$

[1]Following **??**, **??**, we may compute the (characteristic functions associated to the) domain and range of a relation using the following colimit formulas:

$$\begin{aligned} \chi_{\mathrm{dom}(R)}(a) &\cong \operatorname*{colim}_{b \in B}\left(R^b_a\right) && (a \in A) \\ &\cong \bigvee_{b \in B} R^b_a, \\ \chi_{\mathrm{range}(R)}(b) &\cong \operatorname*{colim}_{a \in A}\left(R^b_a\right) && (b \in B) \\ &\cong \bigvee_{a \in A} R^b_a, \end{aligned}$$

where the join $\bigvee$ is taken in the poset $(\{\mathsf{true}, \mathsf{false}\}, \preceq)$ of Constructions With Sets, Definition 3.2.2.1.3.

[2]Viewing $R$ as a function $R\colon A \to \mathcal{P}(B)$, we have

$$\begin{aligned} \mathrm{dom}(R) &\cong \operatorname*{colim}_{y \in Y}(R(y)) \\ &\cong \bigcup_{y \in Y} R(y), \\ \mathrm{range}(R) &\cong \operatorname*{colim}_{x \in X}(R(x)) \\ &\cong \bigcup_{x \in X} R(x), \end{aligned}$$

## 9.3.2 Binary Unions of Relations

Let $A$ and $B$ be sets and let $R$ and $S$ be relations from $A$ to $B$.

**Definition 9.3.2.1.1 ▶ Binary Unions of Relations**

The **union of** $R$ **and** $S$[1] is the relation $R \cup S$ from $A$ to $B$ defined as follows:

- Viewing relations from $A$ to $B$ as subsets of $A \times B$, we define[2]

$$R \cup S \stackrel{\text{def}}{=} \{(a, b) \in B \times A \mid \text{we have } a \sim_R b \text{ or } a \sim_S b\}.$$

- Viewing relations from $A$ to $B$ as functions $A \to \mathcal{P}(B)$, we define

$$[R \cup S](a) \stackrel{\text{def}}{=} R(a) \cup S(a)$$

for each $a \in A$.

[1] *Further Terminology:* Also called the **binary union of $R$ and $S$**, for emphasis.
[2] This is the same as the union of $R$ and $S$ as subsets of $A \times B$.

**Proposition 9.3.2.1.2 ▶ Properties of Binary Unions of Relations**

Let $R$, $S$, $R_1$, and $R_2$ be relations from $A$ to $B$, and let $S_1$ and $S_2$ be relations from $B$ to $C$.

1. *Interaction With Converses.* We have

$$(R \cup S)^\dagger = R^\dagger \cup S^\dagger.$$

2. *Interaction With Composition.* We have

$$(S_1 \diamond R_1) \cup (S_2 \diamond R_2) \overset{\text{poss.}}{\neq} (S_1 \cup S_2) \diamond (R_1 \cup R_2).$$

**Proof 9.3.2.1.3 ▶ Proof of Proposition 9.3.2.1.2**

Item 1: Interaction With Converses

Clear.

Item 2: Interaction With Composition

Unwinding the definitions, we see that:

- The condition for $(S_1 \diamond R_1) \cup (S_2 \diamond R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $b \sim_{S_1} c$;

    or
    - $a \sim_{R_2} b$ and $b \sim_{S_2} c$;
- The condition for $(S_1 \cup S_2) \diamond (R_1 \cup R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ or $a \sim_{R_2} b$;

    and

* $b \sim_{S_1} c$ or $b \sim_{S_2} c$.

These two conditions may fail to agree (counterexample omitted), and thus the two resulting relations on $A \times C$ may differ.

### 9.3.3 Unions of Families of Relations

Let $A$ and $B$ be sets and let $\{R_i\}_{i \in I}$ be a family of relations from $A$ to $B$.

**Definition 9.3.3.1.1 ▶ The Union of a Family of Relations**

The **union of the family** $\{R_i\}_{i \in I}$ is the relation $\bigcup_{i \in I} R_i$ from $A$ to $B$ defined as follows:

- Viewing relations from $A$ to $B$ as subsets of $A \times B$, we define[1]

$$\bigcup_{i \in I} R_i \stackrel{\text{def}}{=} \left\{ (a, b) \in (A \times B)^{\times I} \;\middle|\; \begin{array}{l} \text{there exists some } i \in I \\ \text{such that } a \sim_{R_i} b \end{array} \right\}.$$

- Viewing relations from $A$ to $B$ as functions $A \to \mathcal{P}(B)$, we define

$$\left[ \bigcup_{i \in I} R_i \right](a) \stackrel{\text{def}}{=} \bigcup_{i \in I} R_i(a)$$

for each $a \in A$.

[1]This is the same as the union of $\{R_i\}_{i \in I}$ as a collection of subsets of $A \times B$.

**Proposition 9.3.3.1.2 ▶ Properties of Unions of Families of Relations**

Let $A$ and $B$ be sets and let $\{R_i\}_{i \in I}$ be a family of relations from $A$ to $B$.

1. *Interaction With Converses.* We have

$$\left( \bigcup_{i \in I} R_i \right)^{\dagger} = \bigcup_{i \in I} R_i^{\dagger}.$$

**PROOF 9.3.3.1.3 ▶ PROOF OF PROPOSITION 9.3.3.1.2**

Item 1: Interaction With Converses

Clear. ∎

### 9.3.4 Binary Intersections of Relations

Let $A$ and $B$ be sets and let $R$ and $S$ be relations from $A$ to $B$.

**DEFINITION 9.3.4.1.1 ▶ BINARY INTERSECTIONS OF RELATIONS**

The **intersection of $R$ and $S$**[1] is the relation $R \cap S$ from $A$ to $B$ defined as follows:

- Viewing relations from $A$ to $B$ as subsets of $A \times B$, we define[2]

$$R \cap S \stackrel{\text{def}}{=} \{(a, b) \in B \times A \mid \text{we have } a \sim_R b \text{ and } a \sim_S b\}.$$

- Viewing relations from $A$ to $B$ as functions $A \to \mathcal{P}(B)$, we define

$$[R \cap S](a) \stackrel{\text{def}}{=} R(a) \cap S(a)$$

  for each $a \in A$.

[1] *Further Terminology:* Also called the **binary intersection of $R$ and $S$**, for emphasis.

[2] This is the same as the intersection of $R$ and $S$ as subsets of $A \times B$.

**PROPOSITION 9.3.4.1.2 ▶ PROPERTIES OF BINARY INTERSECTIONS OF RELATIONS**

Let $R$, $S$, $R_1$, and $R_2$ be relations from $A$ to $B$, and let $S_1$ and $S_2$ be relations from $B$ to $C$.

1. *Interaction With Converses.* We have

$$(R \cap S)^\dagger = R^\dagger \cap S^\dagger.$$

2. *Interaction With Composition.* We have

$$(S_1 \diamond R_1) \cap (S_2 \diamond R_2) = (S_1 \cap S_2) \diamond (R_1 \cap R_2).$$

**Proof 9.3.4.1.3 ▶ Proof of Proposition 9.3.4.1.2**

**Item 1: Interaction With Converses**

Clear.

**Item 2: Interaction With Composition**

Unwinding the definitions, we see that:

- The condition for $(S_1 \diamond R_1) \cap (S_2 \diamond R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $b \sim_{S_1} c$;

    and
    - $a \sim_{R_2} b$ and $b \sim_{S_2} c$;
- The condition for $(S_1 \cap S_2) \diamond (R_1 \cap R_2)$ is:
  - There exists some $b \in B$ such that:
    - $a \sim_{R_1} b$ and $a \sim_{R_2} b$;

    and
    - $b \sim_{S_1} c$ and $b \sim_{S_2} c$.

These two conditions agree, and thus so do the two resulting relations on $A \times C$. ∎

### 9.3.5 Intersections of Families of Relations

Let $A$ and $B$ be sets and let $\{R_i\}_{i \in I}$ be a family of relations from $A$ to $B$.

**Definition 9.3.5.1.1 ▶ The Intersection of a Family of Relations**

The **intersection of the family** $\{R_i\}_{i \in I}$ is the relation $\bigcup_{i \in I} R_i$ defined as follows:

- Viewing relations from $A$ to $B$ as subsets of $A \times B$, we define[1]

$$\bigcup_{i \in I} R_i \stackrel{\text{def}}{=} \left\{ (a, b) \in (A \times B)^{\times I} \;\middle|\; \begin{array}{l} \text{for each } i \in I, \\ \text{we have } a \sim_{R_i} b \end{array} \right\}.$$

- Viewing relations from $A$ to $B$ as functions $A \to \mathcal{P}(B)$, we define

$$\left[\bigcap_{i\in I} R_i\right](a) \overset{\text{def}}{=} \bigcap_{i\in I} R_i(a)$$

for each $a \in A$.

[1]This is the same as the intersection of $\{R_i\}_{i\in I}$ as a collection of subsets of $A \times B$.

**PROPOSITION 9.3.5.1.2 ▶ PROPERTIES OF INTERSECTIONS OF FAMILIES OF RELATIONS**

Let $A$ and $B$ be sets and let $\{R_i\}_{i\in I}$ be a family of relations from $A$ to $B$.

1. *Interaction With Converses.* We have

$$\left(\bigcap_{i\in I} R_i\right)^{\dagger} = \bigcap_{i\in I} R_i^{\dagger}.$$

**PROOF 9.3.5.1.3 ▶ PROOF OF PROPOSITION 9.3.5.1.2**

Item 1: Interaction With Converses

Clear. ∎

## 9.3.6 Binary Products of Relations

Let $A$, $B$, $X$, and $Y$ be sets, let $R\colon A \nrightarrow B$ be a relation from $A$ to $B$, and let $S\colon X \nrightarrow Y$ be a relation from $X$ to $Y$.

**DEFINITION 9.3.6.1.1 ▶ BINARY PRODUCTS OF RELATIONS**

The **product of $R$ and $S$**[1] is the relation $R \times S$ from $A \times X$ to $B \times Y$ defined as follows:

- Viewing relations from $A \times X$ to $B \times Y$ as subsets of $(A \times X) \times (B \times Y)$, we define $R \times S$ as the Cartesian product of $R$ and $S$ as subsets of $A \times X$ and $B \times Y$.[2]

- Viewing relations from $A \times X$ to $B \times Y$ as functions $A \times X \to$

$\mathcal{P}(B \times Y)$, we define $R \times S$ as the composition

$$A \times X \xrightarrow{R \times S} \mathcal{P}(B) \times \mathcal{P}(Y) \overset{\mathcal{P}^{\otimes}_{B,Y}}{\hookrightarrow} \mathcal{P}(B \times Y)$$

in $\mathsf{Sets}$, i.e. by

$$[R \times S](a, x) \overset{\text{def}}{=} R(a) \times S(x)$$

for each $(a, x) \in A \times X$.

[1] *Further Terminology:* Also called the **binary product of** $R$ **and** $S$, for emphasis.

[2] That is, $R \times S$ is the relation given by declaring $(a, x) \sim_{R \times S} (b, y)$ iff $a \sim_R b$ and $x \sim_S y$.

**Proposition 9.3.6.1.2 ▶ Properties of Binary Products of Relations**

Let $A$, $B$, $X$, and $Y$ be sets.

1. *Interaction With Converses.* Let

$$R \colon A \nrightarrow A,$$
$$S \colon X \nrightarrow X$$

We have

$$(R \times S)^{\dagger} = R^{\dagger} \times S^{\dagger}.$$

2. *Interaction With Composition.* Let

$$R_1 \colon A \nrightarrow B,$$
$$S_1 \colon B \nrightarrow C,$$
$$R_2 \colon X \nrightarrow Y,$$
$$S_2 \colon Y \nrightarrow Z$$

be relations. We have

$$(S_1 \diamond R_1) \times (S_2 \diamond R_2) = (S_1 \times S_2) \diamond (R_1 \times R_2).$$

**Proof 9.3.6.1.3 ▶ Proof of Proposition 9.3.6.1.2**

**Item 1: Interaction With Converses**

Unwinding the definitions, we see that:

- We have $(a, x) \sim_{(R \times S)^\dagger} (b, y)$ iff:
  - We have $(b, y) \sim_{R \times S} (a, x)$, i.e. iff:
    - We have $b \sim_R a$;
    - We have $y \sim_S x$;
- We have $(a, x) \sim_{R^\dagger \times S^\dagger} (b, y)$ iff:
  - We have $a \sim_{R^\dagger} b$ and $x \sim_{S^\dagger} y$, i.e. iff:
    - We have $b \sim_R a$;
    - We have $y \sim_S x$.

These two conditions agree, and thus the two resulting relations on $A \times X$ are equal.

**Item 2: Interaction With Composition**

Unwinding the definitions, we see that:

- We have $(a, x) \sim_{(S_1 \diamond R_1) \times (S_2 \diamond R_2)} (c, z)$ iff:
  - We have $a \sim_{S_1 \diamond R_1} c$ and $x \sim_{S_2 \diamond R_2} z$, i.e. iff:
    - There exists some $b \in B$ such that $a \sim_{R_1} b$ and $b \sim_{S_1} c$;
    - There exists some $y \in Y$ such that $x \sim_{R_2} y$ and $y \sim_{S_2} z$;
- We have $(a, x) \sim_{(S_1 \times S_2) \diamond (R_1 \times R_2)} (c, z)$ iff:
  - There exists some $(b, y) \in B \times Y$ such that $(a, x) \sim_{R_1 \times R_2} (b, y)$ and $(b, y) \sim_{S_1 \times S_2} (c, z)$, i.e. such that:
    - We have $a \sim_{R_1} b$ and $x \sim_{R_2} y$;
    - We have $b \sim_{S_1} c$ and $y \sim_{S_2} z$.

These two conditions agree, and thus the two resulting relations from $A \times X$ to $C \times Z$ are equal. ∎

### 9.3.7 Products of Families of Relations

Let $\{A_i\}_{i\in I}$ and $\{B_i\}_{i\in I}$ be families of sets, and let $\{R_i\colon A_i \nrightarrow B_i\}_{i\in I}$ be a family of relations.

**Definition 9.3.7.1.1 ▶ The Product of a Family of Relations**

The **product of the family** $\{R_i\}_{i\in I}$ is the relation $\prod_{i\in I} R_i$ from $\prod_{i\in I} A_i$ to $\prod_{i\in I} B_i$ defined as follows:

- Viewing relations as subsets, we define $\prod_{i\in I} R_i$ as its product as a family of sets, i.e. we have

$$\prod_{i\in I} R_i \stackrel{\text{def}}{=} \left\{ (a_i, b_i)_{i\in I} \in \prod_{i\in I}(A_i \times B_i) \;\middle|\; \begin{array}{l} \text{for each } i \in I, \\ \text{we have } a_i \sim_{R_i} b_i \end{array} \right\}.$$

- Viewing relations as functions to powersets, we define

$$\left[\prod_{i\in I} R_i\right]\left((a_i)_{i\in I}\right) \stackrel{\text{def}}{=} \prod_{i\in I} R_i(a_i)$$

  for each $(a_i)_{i\in I} \in \prod_{i\in I} R_i$.

### 9.3.8 The Collage of a Relation

Let $A$ and $B$ be sets and let $R\colon A \nrightarrow B$ be a relation from $A$ to $B$.

**Definition 9.3.8.1.1 ▶ The Collage of a Relation**

The **collage of** $R$[1] is the poset $\mathbf{Coll}(R) \stackrel{\text{def}}{=} \left(\mathrm{Coll}(R), \preceq_{\mathbf{Coll}(R)}\right)$ consisting of:

- *The Underlying Set.* The set $\mathrm{Coll}(R)$ defined by

$$\mathrm{Coll}(R) \stackrel{\text{def}}{=} A \coprod B.$$

- *The Partial Order.* The partial order

$$\preceq_{\mathbf{Coll}(R)}\colon \mathrm{Coll}(R) \times \mathrm{Coll}(R) \to \{\mathsf{true}, \mathsf{false}\}$$

on $\mathrm{Coll}(R)$ defined by

$$\preceq(a,b) \stackrel{\text{def}}{=} \begin{cases} \mathsf{true} & \text{if } a = b \text{ or } a \sim_R b, \\ \mathsf{false} & \text{otherwise.} \end{cases}$$

[1] *Further Terminology:* Also called the **cograph of** $R$.

**NOTATION 9.3.8.1.2 ▶ NOTATION: $\mathsf{Pos}_{/\Delta^1}(A,B)$**

We write $\mathsf{Pos}_{/\Delta^1}(A,B)$ for the category defined as the pullback

$$\mathsf{Pos}_{/\Delta^1}(A,B) \stackrel{\text{def}}{=} \mathsf{pt} \underset{[A],\mathsf{Pos},\mathrm{ev}_0}{\times} \mathsf{Pos}_{/\Delta^1} \underset{\mathrm{ev}_1,\mathsf{Pos},[B]}{\times} \mathsf{pt},$$

as in the diagram

$$\begin{tikzcd}
 & & \mathbf{Pos}_{/\Delta^1}(A,B) \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & & \\
 & \mathbf{Pos}_{/\Delta^1} \underset{\mathbf{Pos}}{\times} \mathsf{pt} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & & \mathsf{pt} \underset{\mathbf{Pos}}{\times} \mathbf{Pos}_{/\Delta^1} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\lrcorner"] & \\
\mathsf{pt} \arrow[dr, "{[A]}"'] & & \mathbf{Pos}_{/\Delta^1} \arrow[dl, "\mathrm{ev}_{[0]}"] \arrow[dr, "\mathrm{ev}_{[1]}"'] & & \mathsf{pt}. \arrow[dl, "{[B]}"] \\
 & \mathbf{Pos} & & \mathbf{Pos} &
\end{tikzcd}$$

**REMARK 9.3.8.1.3 ▶ UNWINDING NOTATION 9.3.8.1.2**

In detail, $\mathsf{Pos}_{/\Delta^1}(A,B)$ is the category where:

- *Objects.* An object of $\mathsf{Pos}_{/\Delta^1}(A,B)$ is a pair $(X, \phi_X)$ consisting of
  - A poset $X$;
  - A morphism $\phi_X \colon X \to \Delta^1$;

  such that we have

$$\phi_X^{-1}(0) = A,$$
$$\phi_X^{-1}(1) = B.$$

- *Morphisms.* A morphism of $\mathsf{Pos}_{/\Delta^1}(A, B)$ from $(X, \phi_X)$ to $(Y, \phi_Y)$ is a morphism of posets $f\colon X \to Y$ making the diagram

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "\phi_X"'] & & Y \arrow[dl, "\phi_Y"] \\ & \Delta^1 & \end{tikzcd}$$

commute.

**PROPOSITION 9.3.8.1.4 ▶ PROPERTIES OF COLLAGES OF RELATIONS**

Let $A$ and $B$ be sets and let $R\colon A \nrightarrow B$ be a relation from $A$ to $B$.

1. *Functoriality.* The assignment $R \mapsto \mathbf{Coll}(R)$ defines a functor

$$\mathbf{Coll}\colon \mathbf{Rel}(A, B) \to \mathsf{Pos}_{/\Delta^1}(A, B),$$

where

- *Action on Objects.* For each $R \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, we have

$$[\mathbf{Coll}](R) \stackrel{\text{def}}{=} (\mathbf{Coll}(R), \phi_R)$$

for each $R \in \mathbf{Rel}(A, B)$, where

  - The poset $\mathbf{Coll}(R)$ is the collage of $R$ of Definition 9.3.8.1.1.
  - The morphism $\phi_R\colon \mathbf{Coll}(R) \to \Delta^1$ is given by

$$\phi_R(x) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } x \in A, \\ 1 & \text{if } x \in B \end{cases}$$

for each $x \in \mathbf{Coll}(R)$.

- *Action on Morphisms.* For each $R, S \in \mathrm{Obj}(\mathbf{Rel}(A, B))$, the action on Hom-sets

$$\mathbf{Coll}_{R,S}\colon \mathrm{Hom}_{\mathbf{Rel}(A,B)}(R, S) \to \mathsf{Pos}(\mathbf{Coll}(R), \mathbf{Coll}(S))$$

of $\mathbf{Coll}$ at $(R, S)$ is given by sending an inclusion

$$\iota\colon R \subset S$$

to the morphism

$$\mathbf{Coll}(\iota)\colon \mathbf{Coll}(R) \to \mathbf{Coll}(S)$$

of posets over $\Delta^1$ defined by

$$[\mathbf{Coll}(\iota)](x) \overset{\text{def}}{=} x$$

for each $x \in \mathbf{Coll}(R)$.[1]

2. *Equivalence.* The functor of Item 1 is an equivalence of categories.

[1]Note that this is indeed a morphism of posets: if $x \preceq_{\mathbf{Coll}(R)} y$, then $x = y$ or $x \sim_R y$, so we have either $x = y$ or $x \sim_S y$ (as $R \subset S$), and thus $x \preceq_{\mathbf{Coll}(S)} y$.

**Proof 9.3.8.1.5 ▶ Proof of Proposition 9.3.8.1.4**

Item 1: Functoriality

Clear.

??: Equivalence

Omitted. ∎

# Appendices

## A Other Chapters

**Preliminaries**

1. Introduction
2. A Guide to the Literature

**Sets**

3. Sets
4. Constructions With Sets
5. Monoidal Structures on the Category of Sets
6. Pointed Sets
7. Tensor Products of Pointed Sets

**Relations**

8. Relations
9. Constructions With Relations
10. Conditions on Relations

**Categories**

11. Categories

**Monoidal Categories**

12. Constructions With Monoidal Categories

**Bicategories**

13. Types of Morphisms in Bicategories

**Extra Part**

14. Notes